# The Optimal Universal Constant in the Nuclear Corner-Tail Estimate Is 3/2

Candidate full solution packet — `fa_banach_001`, lane 3

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**Status.** Candidate full solution, likely valid. The source constant 2 is improved to 3/2, and 3/2 is proved optimal as a constant uniform in $p$ and $q$. Human review is recommended.

## 1 Source question

Hamhalter and Kalenda study bounded subsets of $\mathcal{N}(\ell^q(\Lambda), \ell^p(J))$, the nuclear operators from $\ell^q(\Lambda)$ to $\ell^p(J)$, with $p, q \in (1, \infty)$ [1]. If $P_C$ and $Q_D$ are the canonical coordinate projections for finite $C \subset J$ and $D \subset \Lambda$, define

$$\mathfrak{t}_{p,q}(A) := \inf_{C,D \text{ finite}} \sup_{T \in A} \|T - P_C T Q_D\|_N. \tag{1}$$

Their Theorem 2.1(c), page 5, proves

$$\chi(A) \le \mathfrak{t}_{p,q}(A) \le 2\chi(A),$$

where $\chi$ is the Hausdorff measure of norm noncompactness, and then asks whether the constant 2 is optimal.

## 2 Answer and proof intuition

**Answer.** The constant 2 is not optimal. It can be replaced by 3/2, and 3/2 is the best constant valid simultaneously for all $p, q \in (1, \infty)$.

### Proof intuition

The source proof estimates the complement of the corner projection $T \mapsto P_C T Q_D$ by the generic bound $1 + 1 = 2$. Coordinate projections have extra symmetry. The operators $U = 2P_C - I$ and $V = 2Q_D - I$ are sign-change isometries, and averaging their four actions gives the exact formula

$$T - P_C T Q_D = \frac{1}{4}(3T - UT - TV - UTV).$$

The sum of the absolute coefficients is 3/2, giving the improved bound.

Sharpness is a finite-dimensional projection phenomenon. In the trace-dual operator space, deleting one corner sends a $2 \times 2$ matrix of $\ell^\infty \to \ell^1$ norm 2 to one of norm 3. Finite-dimensional continuity moves this ratio inside the allowed range $p, q \in (1, \infty)$. A replication argument then turns a kernel vector for the corner projection into an infinite bounded set whose ratio $\mathfrak{t}(A)/\chi(A)$ is the same projection-quotient ratio.

$X$ to $Y$ equipped with the nuclear norm is denoted by $N(X, Y)$. The main result on nuclear operators is the following theorem.

**Theorem 2.1.** *Let $J$ and $\Lambda$ be infinite sets and $p, q \in (1, \infty)$.*

(a) *If $p > q$, then $N(\ell^q(\Lambda), \ell^p(J))$ is reflexive and hence all the measures of weak non-compactness vanish for any bounded subset of $N(\ell^q(\Lambda), \ell^p(J))$.*

(b) *If $p \leq q$, then $Z = N(\ell^q(\Lambda), \ell^p(J))$ is not reflexive and for any bounded set $A \subset Z$ we have*

$$\omega(A) = \mathrm{wk}_Z(A) = \mathrm{wck}_Z(A)$$
$$= \inf \left\{ \sup_{T \in A} \|(I - P_C)T(I - Q_D)\|_N \,;\, C \subset J, D \subset \Lambda \text{ finite} \right\},$$

*where $P_C : \ell^p(J) \to \ell^p(J)$ is the canonical projection annihilating coordinates outside $C$ and $Q_D$ is the analogous projection on $\ell^q(\Lambda)$.*

(c) *For any bounded set $A \subset N(\ell^q(\Lambda), \ell^p(J))$ we have*

$$\chi(A) \leq \inf \left\{ \sup_{T \in A} \|T - P_C T Q_D\|_N \,;\, C \subset J, D \subset \Lambda \text{ finite} \right\} \leq 2\chi(A).$$

This theorem will be proved in the next section. The assertion (b) can be viewed as a non-commutative version of Theorem B, as the nuclear operators are, in a sense, a noncommutative version of $\ell^1(\Gamma)$ (this is illustrated by Lemma 3.5(b) below). The assertion (c) is easy and is included for the sake of completeness. It also illustrates the difference between the non-commutative and commutative cases. Indeed, unlike the space of nuclear operators, the space $\ell^1(\Gamma)$ has the Schur property (hence $\omega = \chi$ in this space). The comparison of the formulas in (b) and (c) reveals the different nature of the non-commutative case. Furhter, the assertion (c) does not provide a precise formula for $\chi$, only the two inequlaties. This pair of inequalities cannot be easily replaced by an equality by Example 3.10. We do not know whether the constant 2 is optimal.

The second main result deals with atomic von Neumann algebras. Recall that a *von Neumann algebra* is a $*$-subalgebra $M \subset L(H)$ of the space of bounded linear

Figure 1: Theorem 2.1(c) and the open constant question in arXiv:1711.08906v1, page 5.

## 3 The improved inequality

**Theorem 1.** *Let $J, \Lambda$ be infinite sets, $p, q \in (1, \infty)$, and let $A \subset \mathcal{N}(\ell^q(\Lambda), \ell^p(J))$ be bounded. Then*

$$\chi(A) \leq \mathfrak{t}_{p,q}(A) \leq \frac{3}{2}\chi(A). \tag{2}$$

*Moreover, $3/2$ is the least constant for which the second inequality in (2) holds for all choices of $p, q, J, \Lambda, A$.*

**Lemma 2** (corner-complement norm)**.** *For arbitrary coordinate projections $P = P_C$ on $\ell^p(J)$ and $Q = Q_D$ on $\ell^q(\Lambda)$, the operator*

$$\Psi_{C,D} : \mathcal{N}(\ell^q(\Lambda), \ell^p(J)) \longrightarrow \mathcal{N}(\ell^q(\Lambda), \ell^p(J)), \qquad \Psi_{C,D}(T) = T - PTQ,$$

*satisfies $\|\Psi_{C,D}\| \leq 3/2$.*

*Proof.* Set $U = 2P - I$ and $V = 2Q - I$. These are surjective linear isometries on the respective $\ell^p$- and $\ell^q$-spaces. Left or right composition of a nuclear operator by a surjective isometry preserves its nuclear norm: one inequality follows by composing a nuclear representation, and the reverse inequality follows by composing with the inverse isometry. Since

$$PTQ = \frac{1}{4}(I+U)T(I+V) = \frac{1}{4}(T+UT+TV+UTV),$$

we have

$$\Psi_{C,D}(T) = \frac{1}{4}(3T-UT-TV-UTV).$$

The triangle inequality and invariance of the nuclear norm under the three sign changes give

$$\|\Psi_{C,D}(T)\|_N \le \frac{1}{4}(3+1+1+1)\|T\|_N = \frac{3}{2}\|T\|_N. \qquad \square$$

*Proof of the inequalities in Theorem 1.* For fixed finite $C, D$, the set $\{P_C T Q_D : T \in A\}$ is bounded in a finite-dimensional space. It is therefore relatively compact and lies uniformly within $\sup_{T\in A}\|T - P_C T Q_D\|_N$ of $A$. Taking the infimum gives $\chi(A) \le \mathfrak{t}_{p,q}(A)$.

Fix $\varepsilon > 0$. Choose a finite set $F$ in the nuclear-operator space such that every $T \in A$ has some $S \in F$ with

$$\|T - S\|_N < \chi(A) + \varepsilon.$$

Finite coordinate truncations converge in nuclear norm, so common finite sets $C \subset J$, $D \subset \Lambda$ may be chosen with $\|S - P_C S Q_D\|_N < \varepsilon$ for all $S \in F$. For $T \in A$, choose such an $S$. Lemma 2 yields

$$\begin{aligned}\|T - P_C T Q_D\|_N &\le \|\Psi_{C,D}(T-S)\|_N + \|S - P_C S Q_D\|_N \\ &< \frac{3}{2}(\chi(A)+\varepsilon) + \varepsilon.\end{aligned}$$

Taking the supremum over $A$, then the infimum over $C, D$, and finally letting $\varepsilon \downarrow 0$, proves the upper inequality. $\square$

# 4 Sharpness of 3/2

We first record a standard elementary fact about projections.

**Lemma 3** (projection as a quotient section)**.** *If $\Phi$ is a bounded projection on a normed space $X$ and $\Psi = I - \Phi$, then*

$$\|\Psi\| = \sup_{0\neq b\in\ker\Phi} \frac{\|b\|}{\operatorname{dist}(b, \operatorname{ran}\Phi)}. \tag{3}$$

*In finite dimension the supremum is attained.*

*Proof.* If $x \in X$ and $b = \Psi x$, then $-\Phi x \in \operatorname{ran}\Phi$ and $b - (-\Phi x) = x$, so $\operatorname{dist}(b, \operatorname{ran}\Phi) \le \|x\|$. This gives the inequality "$\ge$" in (3). Conversely, for $b \in \ker\Phi$ and $e \in \operatorname{ran}\Phi$, $\Psi(b-e) = b$, so $\|b\| \le \|\Psi\|\|b-e\|$. Taking the infimum over $e$ gives the reverse inequality. Compactness of the finite-dimensional quotient unit sphere gives attainment. $\square$

For $p, q \in (1,\infty)$, put $Z_{p,q} = \mathcal{N}(\ell_2^q, \ell_2^p)$, and let $\Phi_{p,q}$ keep only the $(1,1)$ matrix entry. By the trace duality used in the source paper [1, Proposition 3.1(a)],

$$Z_{p,q}^* = \mathcal{L}(\ell_2^p, \ell_2^q)$$

isometrically. The adjoint of $\Phi_{p,q}$ is the same $(1,1)$-corner projection, hence the adjoint of $\Psi_{p,q} = I - \Phi_{p,q}$ deletes that entry.

Consider

$$H = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}, \qquad H_0 = \begin{pmatrix} 0 & -1 \\ -1 & -1 \end{pmatrix}.$$

At the finite-dimensional endpoint $p=\infty, q=1$,

$$\|H\|_{\ell_2^\infty \to \ell_2^1} = \max_{|x_1|,|x_2|\le 1} (|x_1-x_2|+|x_1+x_2|) = 2,$$
$$\|H_0\|_{\ell_2^\infty \to \ell_2^1} = \max_{|x_1|,|x_2|\le 1} (|x_2|+|x_1+x_2|) = 3.$$

The norms of a fixed $2\times 2$ matrix from $\ell_2^p$ to $\ell_2^q$ depend continuously on $(p,q) \in [1,\infty]^2$. Thus, for example with $p_m = m$ and $q_m = 1+1/m$,

$$\|\Psi_{p_m,q_m}\| = \|\Psi^*_{p_m,q_m}\| \ge \frac{\|H_0\|_{\ell_2^{p_m}\to\ell_2^{q_m}}}{\|H\|_{\ell_2^{p_m}\to\ell_2^{q_m}}} \longrightarrow \frac{3}{2}. \tag{4}$$

It remains to turn these finite-dimensional projection norms into ratios of the two quantities in Theorem 1. Fix $p, q$ and choose, using Lemma 3, a nonzero $b \in \ker \Phi_{p,q}$ such that

$$\frac{\|b\|_N}{\delta} = \|\Psi_{p,q}\|, \qquad \delta := \operatorname{dist}(b, \operatorname{ran}\Phi_{p,q}). \tag{5}$$

For each $n \ge 2$, embed $b$ as a nuclear operator $T_n$ supported on the coordinates $\{1,n\}$ in both $\ell^q(\mathbb{N})$ and $\ell^p(\mathbb{N})$, using coordinate 1 for the first coordinate of $b$. Let $E_{11}x = x_1e_1$, choose $L > \|b\|_N$, and set

$$\mathcal{A}_b = \{T_n : n \ge 2\} \cup \{LE_{11}\}.$$

We claim

$$\chi(\mathcal{A}_b) = \delta, \qquad \mathfrak{t}_{p,q}(\mathcal{A}_b) = \|b\|_N. \tag{6}$$

For the first upper bound, choose an element $e$ of the one-dimensional space $\operatorname{ran}\Phi_{p,q}$ with $\|b-e\|_N = \delta$. Under every embedding it is the same scalar multiple of $E_{11}$, so the two-point set $\{e, LE_{11}\}$ lies within $\delta$ of $\mathcal{A}_b$. Hence $\chi(\mathcal{A}_b) \le \delta$.

Conversely, let $F$ be any finite subset of the infinite nuclear-operator space and let $\varepsilon > 0$. Choose common finite coordinate sets $C, D$, both containing 1, such that $\|S - P_C S Q_D\|_N < \varepsilon$ for every $S \in F$, and choose $n \notin C \cup D$. Compression to the rows and columns $\{1,n\}$ is contractive in nuclear norm. The compression of $P_C S Q_D$ belongs to the $(1,1)$-corner, so the compression of $S$ is within $\varepsilon$ of $\operatorname{ran}\Phi_{p,q}$. Therefore, for every $S \in F$,

$$\|T_n - S\|_N \ge \delta - \varepsilon.$$

This proves $\chi(\mathcal{A}_b) \ge \delta$.

For the tail equality, if either $1 \notin C$ or $1 \notin D$, then the tail of $LE_{11}$ has norm $L > \|b\|_N$. If $1 \in C \cap D$, choose $n \notin C \cup D$. Since $b \in \ker\Phi_{p,q}$, its $(1,1)$ entry is zero, hence $P_C T_n Q_D = 0$, and the tail norm is $\|b\|_N$. This proves the lower bound for $\mathfrak{t}$. For the reverse bound, take increasing finite sets $C = D = \{1,\dots,N\}$: the finitely many $T_n$ supported inside the corner have zero tail, and every remaining $T_n$ has tail norm $\|b\|_N$. Thus (6) holds.

Combining (5), (6), and (4) gives bounded sets with $\mathfrak{t}_{p_m,q_m}(\mathcal{A}_{b_m})/\chi(\mathcal{A}_{b_m}) \to 3/2$. No smaller universal constant can hold. This finishes the proof of Theorem 1.

## 5 Verification, scope, and novelty

The reusable script `code/check_corner_constant.py` checks the sign-flip identity on random matrices and exactly enumerates the four vertices needed for the endpoint norms 2 and 3. It reports endpoint ratio 1.5. These checks are diagnostic only; no numerical statement is used in the proof.

The result determines the constant uniform over all $p, q$. It does not determine the potentially smaller optimal constant for a fixed pair $(p, q)$. It also does not address the source paper's distinct problems on $C(K)$, compact operators, general $C^*$-algebras, or general von Neumann preduals.

The run's registry, solution, attempt, and proof-gap indexes were searched by arXiv id, exact title, "constant 2 optimal," "three halves," "corner," and "nuclear." Web/arXiv searches used the exact source sentence, source title and authors, and close nuclear-operator constant phrases. They found the source paper but no later paper explicitly resolving this question or containing this 3/2 argument. This is a bounded novelty check, not a claim of exhaustive bibliographic certification.

**Human-review recommendation.** Verify especially the orientation of the trace-dual corner map and the two equalities in (6). Both are isolated above so the packet can be audited without reconstructing the discovery process.

## References

[1] J. Hamhalter and O. F. K. Kalenda, *Measures of Weak Non-Compactness in Spaces of Nuclear Operators*, Math. Z. 292 (2019), no. 1–2, 453–471; arXiv:1711.08906v1.